BATYRKHAN KANAPIYA

Java Back-end Developer | Software Engineering Student

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SUMMARY

I'm a second-year Software Engineering student at Astana IT University, and I love building things that actually work. My projects go from a desktop app that controls a computer with face gestures to an AI service that finds violations in bank cash desk videos, which I built for Datathon'26. I have worked with Flask, Django, FastAPI and PostgreSQL, shipped apps with Docker and Railway, and I study Java, algorithms and design patterns at university. Now I'm looking for an internship as a back-end developer, where I can solve real tasks in a team, learn from code review and grow fast.

EDUCATION

Astana IT University, Bachelor's degree in Software Engineering 2025 - present · Astana

2nd year, group SE-2506. Relevant courses: Object-Oriented Programming (Java), Design and Analysis of Algorithms, Software Design Patterns, Operating Systems (Linux), Front-end Development.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, C++, HTML / CSS
Back-end	Spring Boot, FastAPI, Flask, Django, REST / JSON APIs, authentication, ORM (SQLAlchemy, Django ORM), Alembic
Databases	PostgreSQL, Redis, Supabase
Front-end	HTML5, CSS3 (responsive layout, animations), JavaScript (DOM, Fetch API), React
Tools / DevOps	Git & GitHub, GitHub Actions (CI), Docker & Docker Compose, Linux (Ubuntu, Fedora), Gunicorn, Railway, Netlify, PyInstaller, IntelliJ IDEA
AI / ML	Computer vision (OpenCV, MediaPipe), object detection (RT-DETR, Hugging Face Transformers), PyTorch
Concepts	OOP, data structures & algorithms, design patterns, multithreading, background job queues
Soft skills	Fast self-learner, problem solving, responsibility, time management, communication, adaptability, teamwork

PROJECTS

GestureApp | Hands-free PC control with face gestures **MAIN PROJECT** github.com/kanzzzwag/GestureApp

Python · OpenCV · MediaPipe · CustomTkinter · multithreading · PyInstaller

- Desktop app that controls the computer with the face: double blink, left / right wink, mouth open, smile and eyebrow raise trigger keyboard actions.
- Real-time recognition with MediaPipe Face Landmarker; my own eye and mouth aspect-ratio metrics (EAR / MAR), normalized by face size, with thresholds and cooldowns against false triggers.
- Camera loop runs in a separate worker thread with thread-safe frame sharing, so the UI never freezes. Modular architecture (core / gui / utils): live preview, gesture-to-key binding, JSON config, logging, tray mode.
- Packaged as a Windows .exe (PyInstaller) with a full installer (Inno Setup).

Kiye | AI video analysis for bank cash desks (Datathon'26, ForteBank case) github.com/kanzzzwag/Kiye

Python · FastAPI · PostgreSQL · Redis · Docker · RT-DETR · OpenCV

- Service that analyzes video from a camera above a bank cash desk and finds discipline violations: type, timecode, 1-3 evidence frames and the related National Bank rule.
- ML pipeline: RT-DETR object detection, people tracking, scene change analysis and a YAML rule engine for 11 violation types. 89% precision and 73% recall on 42 test videos, fully offline on CPU.
- Back-end: FastAPI REST API with API-key auth and rate limiting, Redis + RQ job queue with a separate worker, PostgreSQL with Alembic migrations, CSV export.
- Docker Compose setup (api, worker, db, redis), pytest tests and GitHub Actions CI (tests, dependency audit, Docker build).

AITU Collab | Team-finding platform for students github.com/kanzzzwag/AITU-Collab

Python · Flask · PostgreSQL · SQLAlchemy · Jinja2 · Gunicorn · Railway

- Full-stack platform where students publish project ideas and find teammates for courses and hackathons.
- Designed 5 relational models (User, Project, JoinRequest, Review, Message) with full CRUD across 14 routes.
- Login with Flask-Login and hashed passwords, join-request workflow (pending / accepted / declined), 1-5 star peer reviews, team chats. Deployed on Railway (Gunicorn + cloud PostgreSQL).

qBattle | Music tournament voting platform github.com/kanzzzwag/qBattle

Django · PostgreSQL · JavaScript · Gunicorn

- Bracket-style tournaments where users vote between tracks; JSON endpoints in Django and Fetch API on the client, so voting works without page reloads.
- Custom Django management commands for data seeding and admin setup, automatic migrations on deploy.

ACTIVITIES, CERTIFICATES & LANGUAGES

- **Datathon'26**: participant (Forte, DataSci, NU Impact Foundation), built the Kiye project
- **AI & Digital Bridge 2026**: Astana: volunteer, 12 hours (Oct 1-3, 2026)
- **GitHub Foundations (GH-900)**: certification in preparation
- **Languages**: Kazakh (native), Russian (fluent), English (intermediate, university studies in English)